

Artificial Intelligence (CS303)

Lecture 4: Principles of Machine Learning

Hints for this lecture

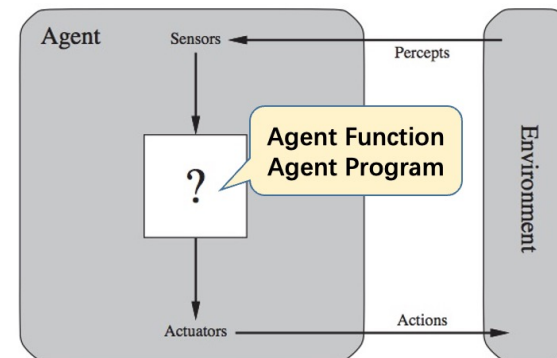
- Learning = Search for hypothesis/functions that generalize well.

Outline of this lecture

- What is Learning
- Key Questions for Learning
- Learning Paradigms and Principles

Why Learning is Important?

- Scientific perspective
 1. Human behavior is not a static program, i.e., we might behave differently as growing up.
 2. Learning is universal and a major source of our behavioral change.
- Engineering perspective
 1. It might be too challenging for an engineer to design an agent function from scratch



What is Learning (in AI)?

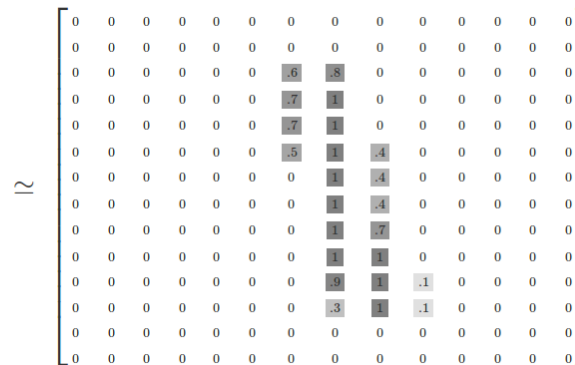
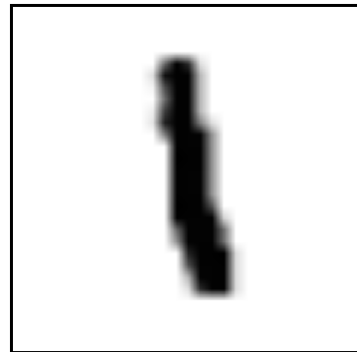
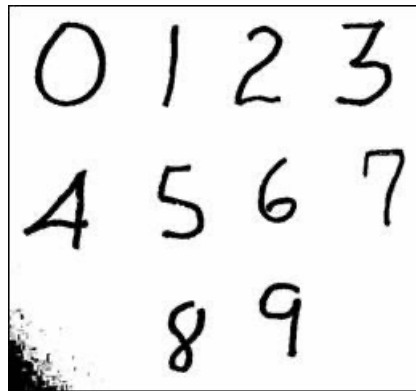
- Examples for human learning
 - Learn similarity between objects
 - Learn to recognize objects
 - Learn to pass the college entrance examination
- **Machine Learning**: Given some observations (data) from the environment, how could an agent improve its agent function?
- Intuitive assumptions
 - the data share something in common
 - “something” could be obtained by an algorithm/program

To Learn What?

- Ideally, the purpose of (Machine) Learning is to achieve
 - a “universal” agent function that can give the appropriate output for any input that we can imagine, or
 - To be realistic, to automatically obtain domain-specific agent functions for specific applications.
- We take the task of learning to recognize a hand-written digit as an illustrative example below, which should be much easier than to learn the to make a good living.

An example

- Hand-written Digit Recognition
 - Data: image of a hand-written digit
 - Agent function: a function mapping **an image to a digit**
 - Improvement: how many images can be recognized correctly.



transform to a vector

A Naive Parametric Method

- Classify a data to the class with the highest posteriori probability

$$P(w_j|x) = \frac{p(x|w_j)P(w_j)}{p(x)}$$

$$P(w_2|x) > P(w_1|x) \quad \longleftrightarrow \quad \ln p(x|w_i) + \ln P(w_i) > \ln p(x|w_i) + \ln P(w_i)$$

- Assumption: data follows independent identically distribution

A Naive Parametric Method

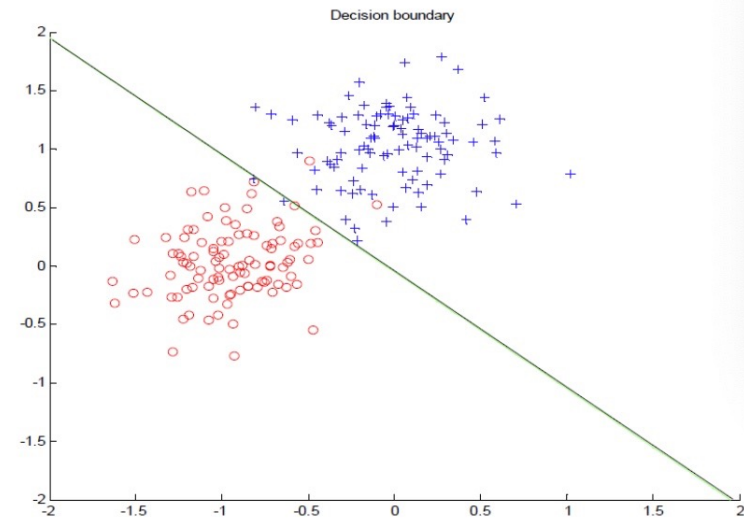
- In addition to the i. i. d assumption, further assume the data follows a specific distribution (e.g., Gaussian distribution)
- Estimate the prior and the likelihood.
- “parametric”: the assumption on the probability density function.
- Parametric methods usually do not involve parameters to fine-tune, while Nonparametric methods usually do.

Agent as A Linear Function

- Find a straight line/hyper-plane to separate datum from different classes.

$$g(\mathbf{x}) = \mathbf{w}^t \mathbf{x} + b$$

input space $\mathcal{X} \subset \mathbb{R}^d$ and output space $\mathcal{Y} \in \{0, 1\}$



Key Questions for Machine Learning

- Given some **observations (data)** from the environment, how could an agent **improve** its **agent function**?
 - What is the format of the data? (data representation)
 - What does the agent function look like? (model representation)
 - How to measure the “improvement”? (objective function)
 - What is the learning algorithm? (to get a good agent function)

Representation + Algorithm + Evaluation = Agent function/Model

Search in a model/hypothesis space

Learning Principles

Generalization, Generalization, Generalization: the learned agent function is expected to be able to handle previously unseen situations. (举一反三, 不要刻舟求剑)

- How to calculate/estimate the generalization?
 - Design an appropriate objective function for learning
- Overfitting: the model perfectly fit the seen data perfectly, but cannot generalize well.
 - Occam's Razor: An explanation of data should be made as simple as possible, but no simpler, i.e., **model complexity** should be controlled.

Learning Paradigms

A Machine Learning process typically involves two phases

- Training: build the agent function
- Testing/Inference: test the agent function/deploy the agent function in real use.

Different ML techniques may use different training/learning paradigms

- **Supervised Learning**: the correct answer is available to the learning algorithm.
- **Reinforcement Learning**: the only feedback is the reward of an output, e.g., the output is correct or not (the correct answer is **not** given).
- **Unsupervised Learning**: no correct answer is available.

Summary

- Machine learning is basically the search for a model/hypothesis/rule that can map an input from the environment to an appropriate output, such that we don't need to program them in advance.
- Different settings of learning may lead to different learning algorithms.
- Generalization is the ultimate goal of learning, but usually uneasy to measure.
- In this course, we mainly use classification problem as an example:
 - Classification is a sufficiently abstract and have numerous applications.
 - It has so far led to many successful applications of machine learning and AI.

What is classification?

- Given a set of **class labels** and a set of **training data**, usually represented by a set of **features**, to achieve a **classifier** that (ideally) can assign the correct label to any previous unseen data.

ID	Height	Weight
John	1.75米	80KG
Mike	1.8米	75KG

- In most literature, classification refers to supervised learning, i.e., labels for the training data are given to the learning algorithm.

To be continued